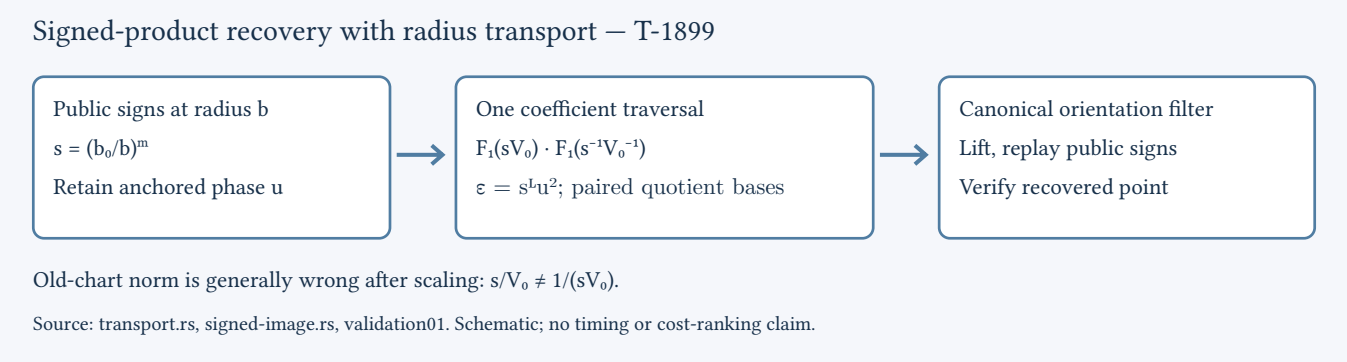
Signed-product recovery across changing radii

T-1899 · Prime-field optional recovery adapter · 2026-11-08

The signed reciprocal equation now runs through canonical phase/orientation state and a public-point recovery adapter. One coefficient traversal evaluates two reciprocal bases per active scaled branch. Existing source queries, endpoint handling, history replay and final point verification are retained.



Why transport matters. At a new radius the variable is sV_0. Conjugation in the old chart maps it to s/V_0, whereas the required inverse is $1/(sV_0)$. The implementation evaluates F_1 at sV_0 and $s^{-1}(W_0-V_0)$, then multiplies. Its equation can contain both old-chart components and is passed to the orientation-preserving filter. At s=1, the ordinary norm is used. Phase updates use $\varepsilon=s^L u^2$.

Exact qualification. The first build and all checks passed: 96 transported quotient products, 3,640 exact image states, and 843 public target/cap cases recovered with 735 subsequent image updates. A negative control records 1,749 wrong-chart products. Repeated radii, zero scalar, ramified and repeated quotient factors are covered. Inherited stdout is byte-identical. These are finite correctness checks, not timing trials. Producer evidence awaits independent signed review.

Formal scope. Lean4.34.1 checks the scaled inverse-pair and transported-trace identities over an arbitrary commutative ring. Axioms are propext, Classical.choice and Quot.sound. The finite-field weighted projection and runtime refinement remain written/finite-check obligations.

Cost. For L coefficients at scale s≠1, the balanced pair evaluator uses 4(L−1)+1 quadratic products including the final product, plus one checked scalar inverse per active branch evaluation. Coefficient transitions, mode0, allocations/copies, canonical filtering, lift, source and replay work remain. A single coefficient traversal is not a single quotient evaluation. Candidate retention and full cost versus mode2 need a frozen matched comparison.

Evidence. transport.rs, generated signed-image.rs and signed-adapter.rs, image-checks.rs, DERIVATION.md, Transport.lean and validation01/receipt.json. No new build/runtime failure, cloud job, production panel or automatic method switch. Historical full-Cargo missing-fixture limitation unchanged.

Sources. Prior signed-reciprocal derivation and existing canonical-image, quadratic and persistent modules; NIST DLMF18.5.1, https://dlmf.nist.gov/18.5.E1. KB unavailable; exact-source fallback recorded.